Lecture 6

Inductor and Transformer

Slide Courtesy:
1.Electronics Technology Fundamentals, 3rd ed. Paynter and Boydell
2. Electronics Fundamentals 9th edition, Floyd/Buchla/Snyder

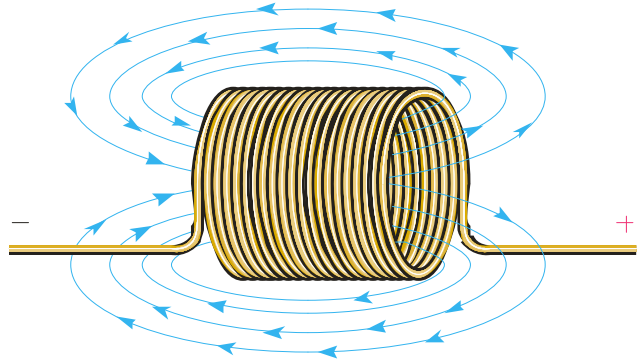
Outline

Topics:

- Basics of Inductor – Self Inductance
- Types of Inductors
- Series and parallel inductors
- Inductors in DC circuits
- Inductors in AC circuits
- Inductor Applications

The Basic Inductor

- When a length of wire is formed into a coil., it becomes a basic inductor.
- When there is current through an inductor, an electromagnetic field is established.
- When the current changes, the electromagnetic field also changes. An increase in current expands the field, and a decrease in current reduces it.

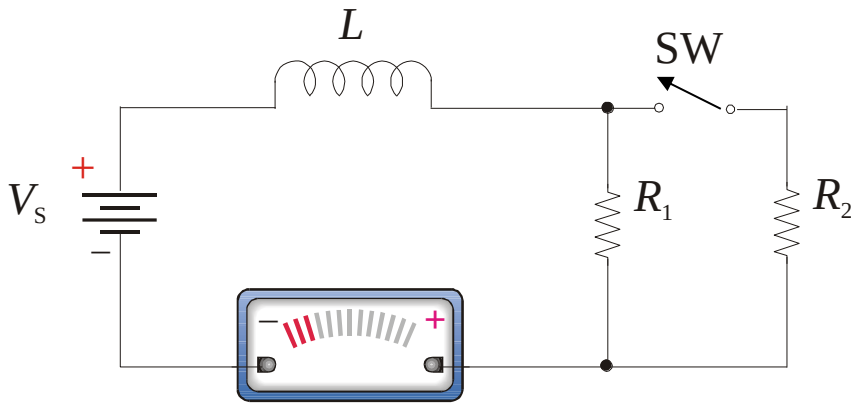


- Its working principle is based on **Faraday's Law of Electromagnetic Induction**, which states that a changing current through a coil induces a voltage (EMF) that opposes the change in current **(Lenz's Law)**.

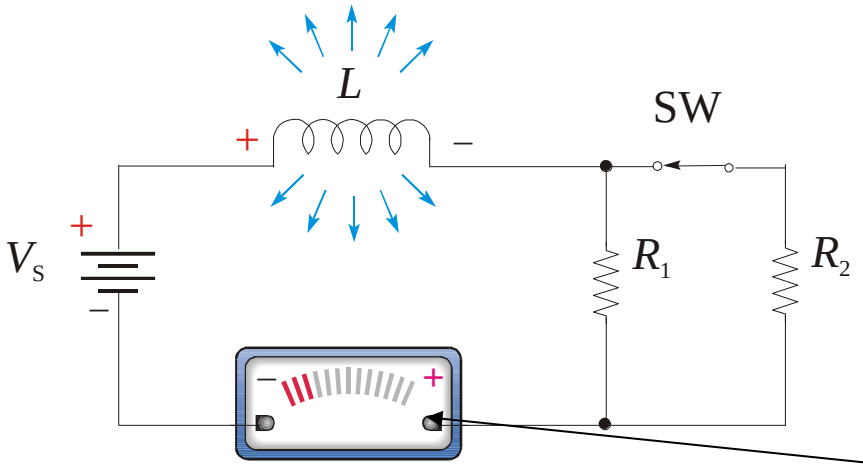
Faraday's Law: The amount of voltage induced in a coil is directly proportional to the rate of change of the magnetic field with respect to the coil.

Lenz's Law: When the current through a coil changes and an induced voltage is created as a result of the changing magnetic field, the direction of the induced voltage is such that it always opposes the change in the current.

A basic circuit to demonstrate Lenz's law

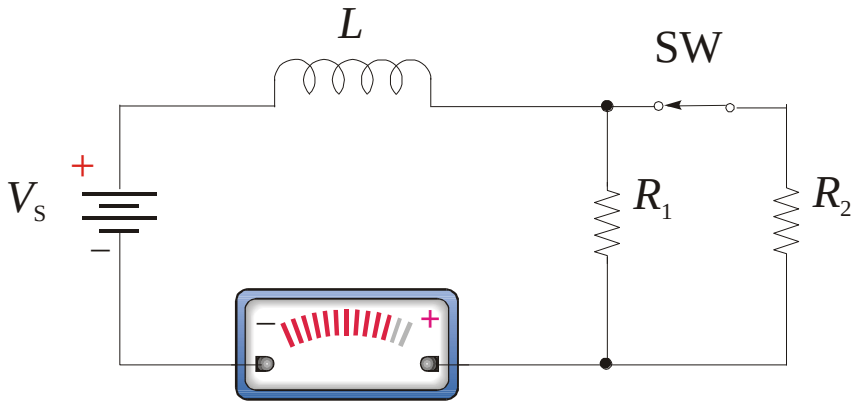


Initially, the SW is open and there is a small current in the circuit through L and R_1



SW closes and immediately a voltage appears across L that tends to oppose any *change* in current.

Later, the meter reads a higher current because of the load change.

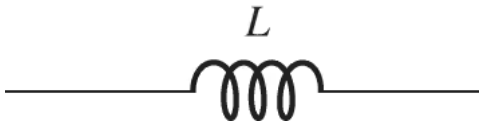


Self-Inductance (L)

- The changing electromagnetic field develops an induced voltage across the coil in a direction to oppose the change in current. This property is called **self-inductance** but is usually referred to as simply inductance, symbolized by L.
- **Inductance (L):** The property of an inductor to oppose changes in current.

Inductor

- **An inductor** is a passive electrical component that stores energy in a magnetic field when electric current flows through it.
- **Unit of Measure** : **Henry (H)**
  - Inductance is measured in volts per rate of change in current
  - When a change of 1A/s induces 1V across an inductor, the amount of inductance is said to be 1 H
- Voltage across an inductor is proportional to the rate of change of current.



$$v_L = L\frac{di}{dt}$$

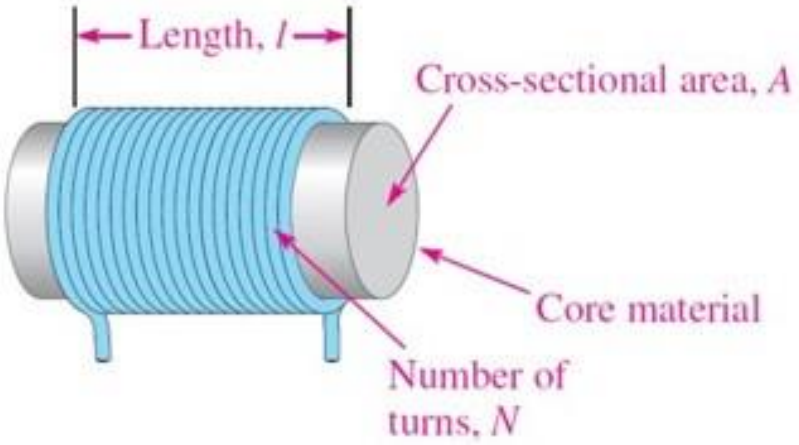
- **Energy Storage :** An inductor stores energy in the magnetic field created by the current. The energy stored is expressed as,

$$\mathrm{E} = \frac{1}{2}LI^2$$

Factors Affecting Inductance

Four factors affect the amount of inductance for a coil.

- Number of turns (N)
- Core material (μ)
- Length of coil
- Cross-sectional area



The equation for the inductance of a coil is

$$L = \frac{N^2 \mu A}{l}$$

where

L = inductance in Henries (H)

N = number of turns of wire

μ = permeability in H/m

l = coil length in meters (m)

Types of Inductors

There are a variety of inductors, depending on the amount of inductance required and the application. Some, with fine wires, are encapsulated and may appear like a resistor. The coil frequently is wound on a ferromagnetic core to increase the inductance.

Common symbols for inductors (coils) are



Air core | Iron core | Ferrite core | Variable

Schematics often use the air core symbol regardless of the actual core material.

Practical Inductors



Encapsulated | Wire wound, high current | Toroid coil | Variable

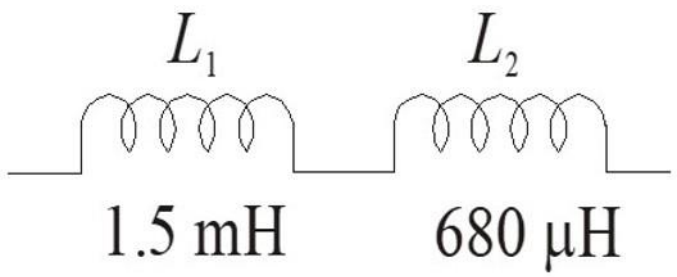
Series Inductors

When inductors are connected in series, the total inductance is the sum of the individual inductors. The general equation for inductors in series is

$$L_T = L_1 + L_2 + L_3 + \ldots L_n$$

Example

If two inductors are connected in series as shown below, the total inductance is 2.18 mH.



Parallel Inductors

When inductors are connected in parallel, the total inductance is smaller than the smallest one. The general equation for inductors in parallel is

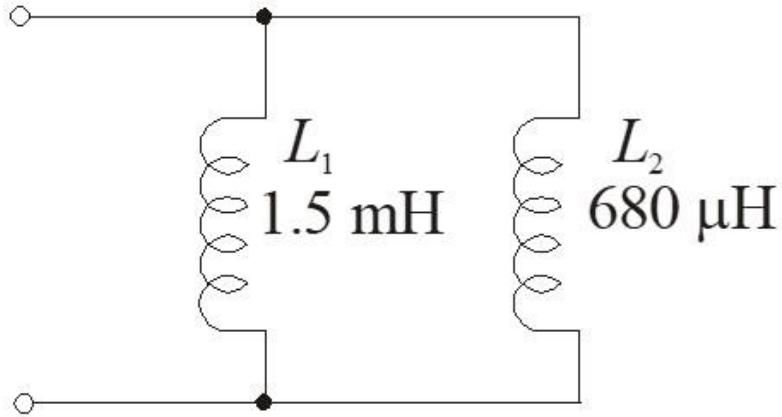
$$L_T = \frac{1}{\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \ldots + \frac{1}{L_T}}$$

The total inductance of two inductors is

$$L_T = \frac{1}{\frac{1}{L_1} + \frac{1}{L_2}}$$

…or you can use the product-over-sum rule.

Q. If a 1.5 mH inductor is connected in parallel with a inductor, the total inductance is 468 µH.

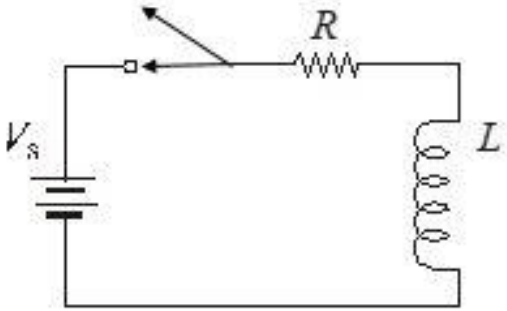


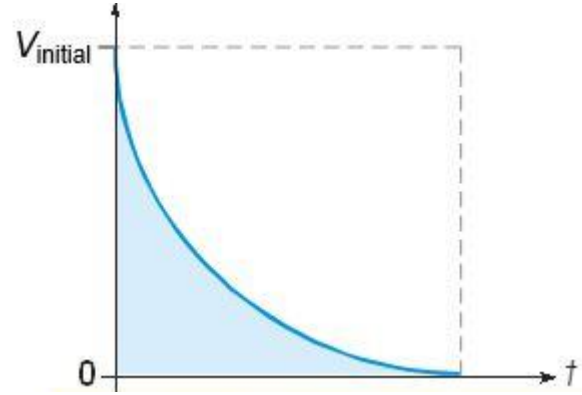
Inductors in DC Circuits

Initial Behavior: Transient State
When a DC voltage is first applied to an inductor, it resists the change in current. The inductor creates an opposing voltage (induced voltage) to prevent sudden current increase.

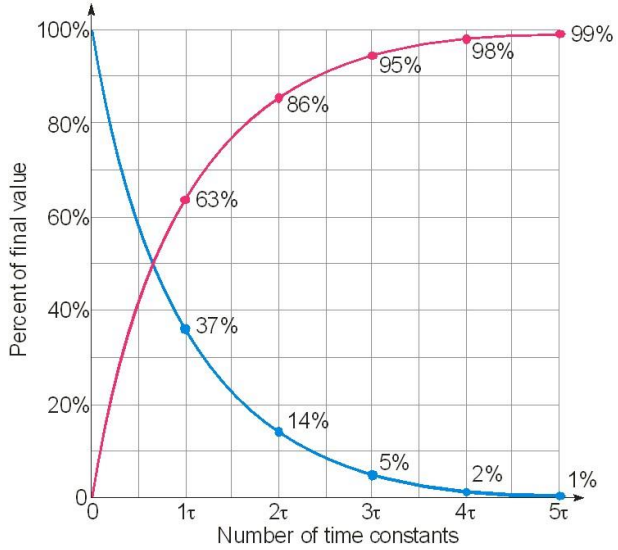
Steady-State Behavior:
Once current is steady, the inductor acts like a **short circuit** (Zero resistance).
Voltage across Inductor: VL=0V (after steady-state is reached).
Current: Constant DC current flows through the inductor without opposition.

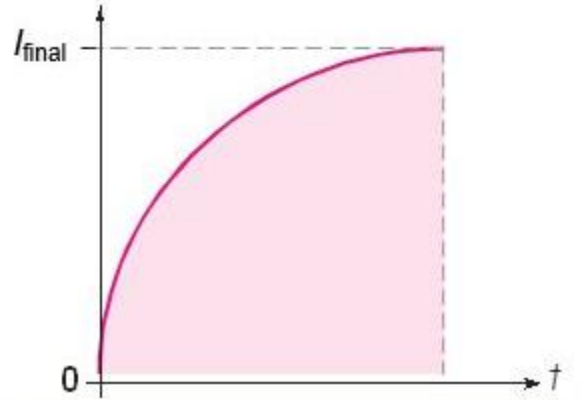
Time constant: $\tau = {}^{L}/_{R}$
The time taken for current to rise to 63% of its final value.





Inductor voltage after switch closure



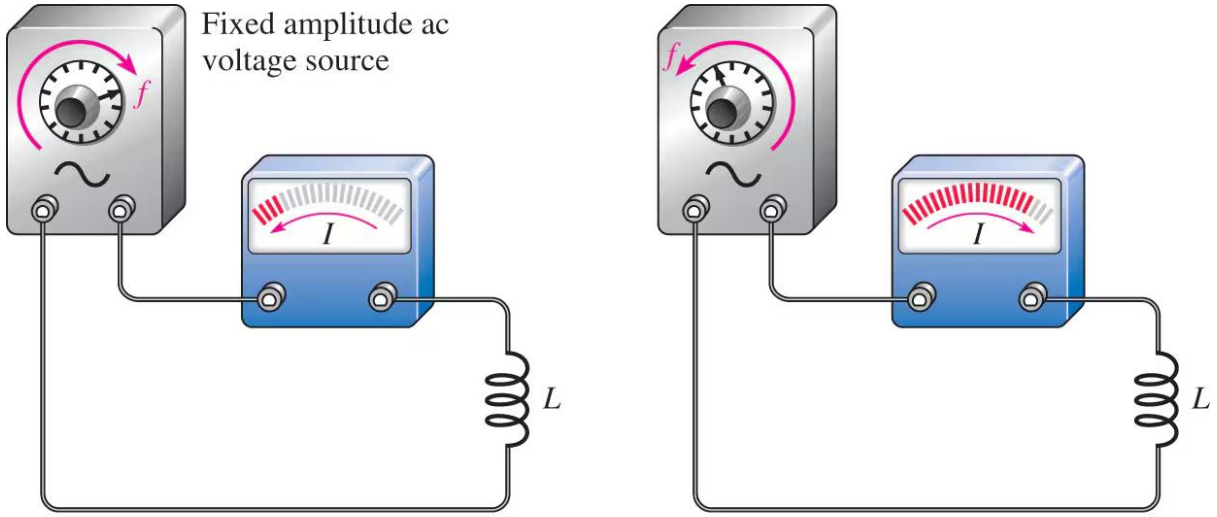


Inductor current after switch closure

Inductors in AC Circuits

In AC circuits, inductor passes ac but with an amount of opposition that depends on the **frequency of the AC**.

- When the frequency of the source voltage increases, its rate of change increases
- This induces more voltage across the inductor that opposes the current and causes it to decrease in amplitude. Similarly, a decrease in frequency will cause an increase in current.



(a) Current decreases when the frequency increases.

(b) Current increases when the frequency decreases.

Inductive reactance (X_L) is the opposition to ac by an inductor.

The equation for inductive reactance is($X_L = 2\pi f L$

Q. What is the reactance of a 33 mH inductor at a frequency of 550 kHz? $114\ \Omega$

Q. What is the inductance of an inductor that has a reactance of 1.57 kΩ at a frequency of 1.25 MHz? $200\ \mu H$

Applications

- **Power Supply Filters (LC Filters) :** Used in power supplies to smooth out AC ripples in DC voltage.
- **Inductive Sensors & Proximity Detection :** Inductors are used in inductive proximity sensors to detect metallic objects without contact.
- **Wireless Charging Systems:** Inductive coupling enables wireless power transfer in smartphone chargers and electric vehicles.

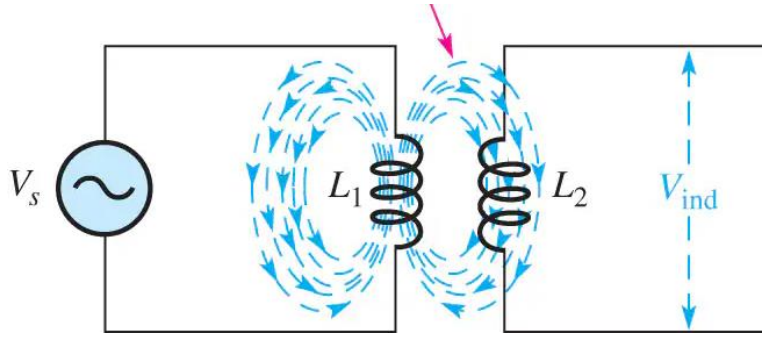
Transformer

Topics:

- Mutual Inductance
- Basic Transformer
- Turns Ratio
- Step-down and Step-up transformer
- Tapped and Multiple-winding Transformer

Mutual Inductance

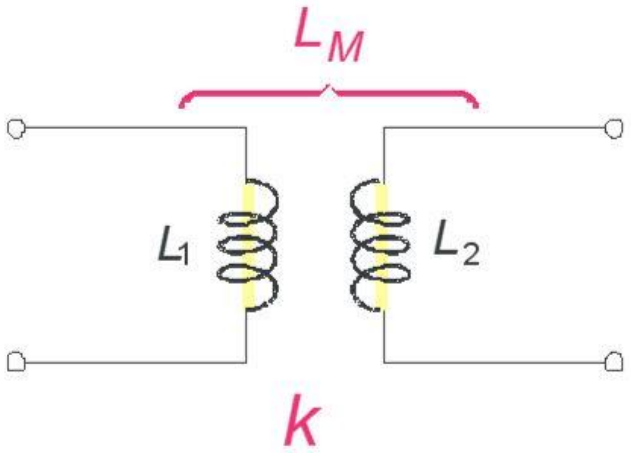
- When two coils are placed close to each other, a changing flux in one coil will cause an induced voltage in the second coil. The coils are said to have **mutual inductance** $(\mathrm{L_M})$



Electrically isolated, Magnetically coupled coils

  - The mutual inductance is established by the inductance of each coil (and) and by the amount of coupling (k) between the two coils.
  - The **coefficient of coupling (k)**is a measure of how well the coils are linked; it is a number between 0 and 1.

If half of the total flux produced by coil 1 links coil 2, then k =0.5
The formula for mutual inductance is

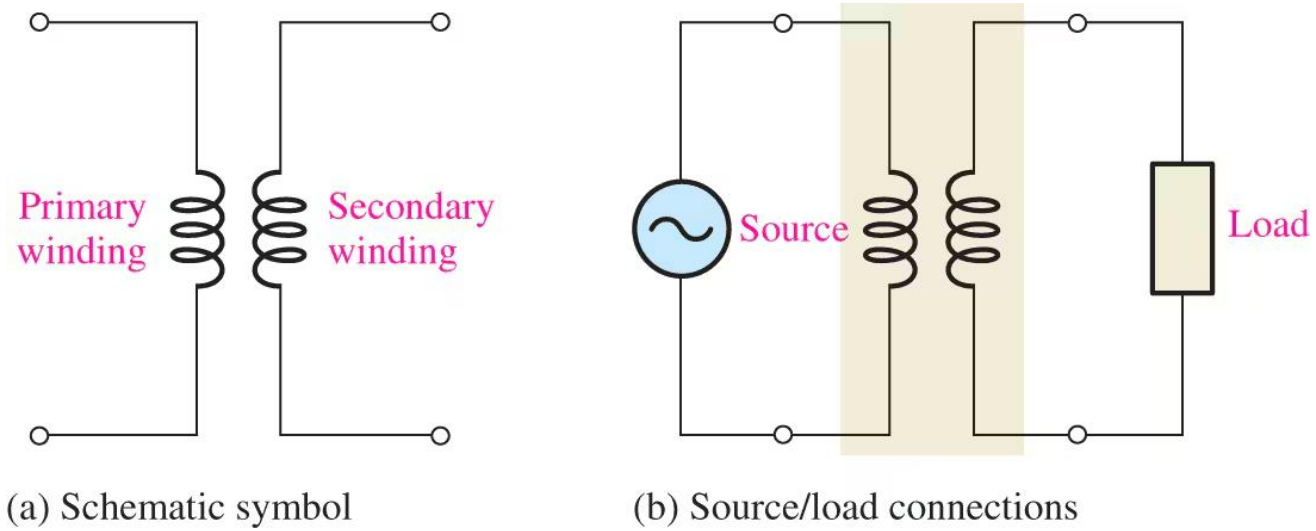


$$L_M = k\sqrt{L_1 L_2}$$

$k =$ the coefficient of coupling (dimensionless)
$L_1, L_2 =$ inductance of each coil (H)

The Basic Transformer

- A transformer is an electrical device constructed of two or more coils of wire (windings) electromagnetically coupled to each other so that there is a mutual inductance for the transfer of power from one winding to the other.
- One coil is called the **primary winding,** and the other coil is called the **secondary winding**, as indicated.



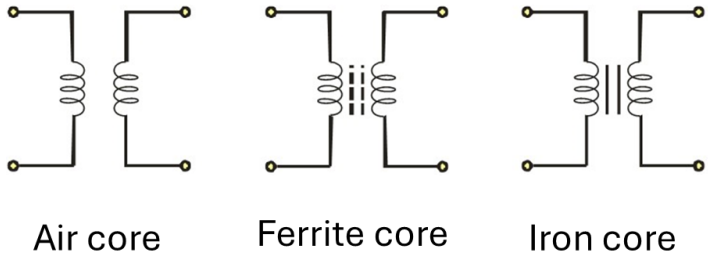
(a) Schematic symbol (b) Source/load connections

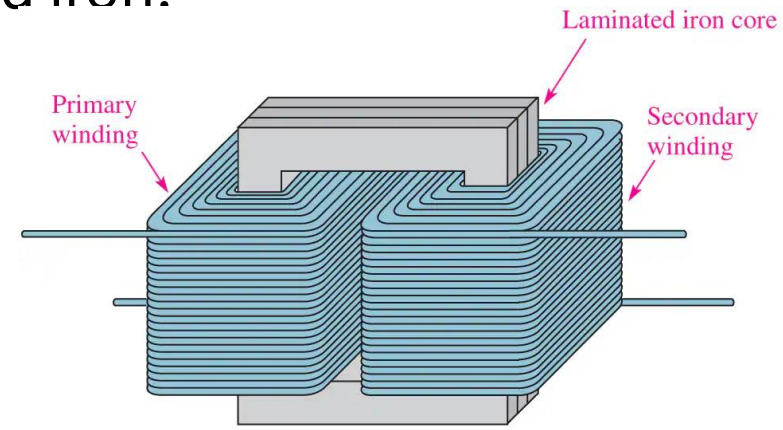
Transformer types

The windings of a transformer are formed around the **core**.

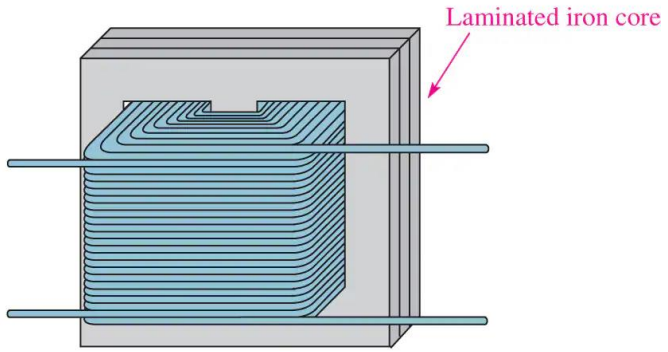
The core provides both a physical structure for placement of the windings and a magnetic path so that the magnetic flux is concentrated close to the coils.

Three general categories of core material are air, ferrite, and iron.

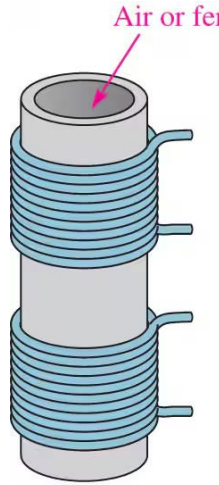




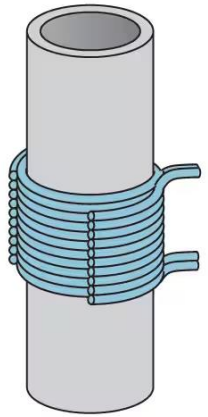
(a) Core type has each winding on a separate leg.



(b) Shell type has both windings on the same leg.



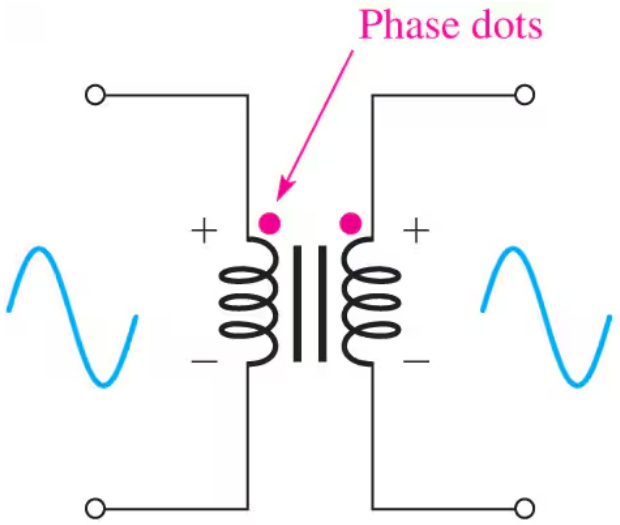
(a) Loosely coupled windings



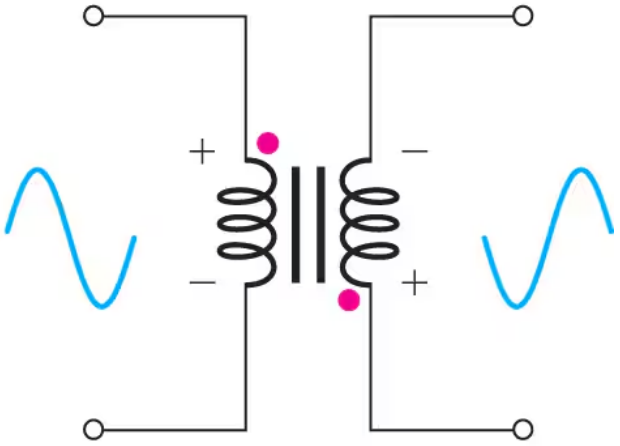
(b) Tightly coupled windings. Cutaway view shows both windings.

Direction of Windings

- The direction of the windings determines the polarity of the voltage across the secondary winding with respect to the voltage across the primary.
- Phase dots are sometimes used to indicate polarities.



(a) Voltages are in phase.

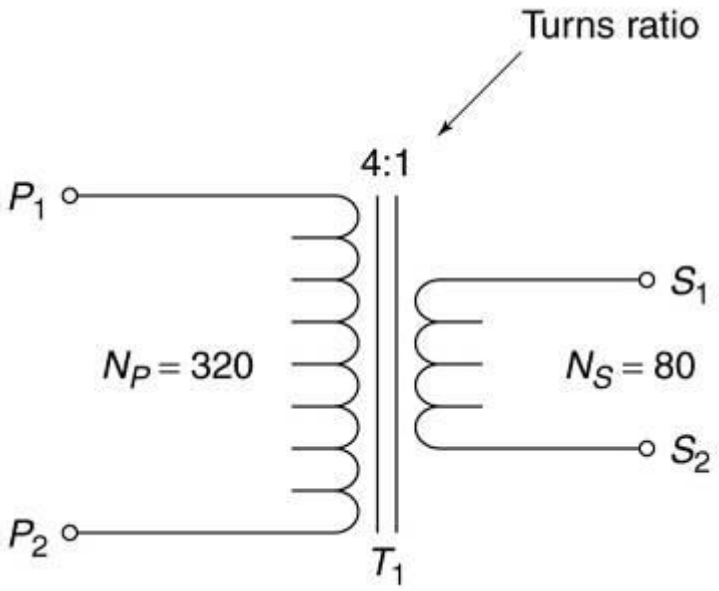


(b) Voltages are out of phase.

Turns Ratio

The turns ratio is defined as the ratio of the number of turns in the primary winding to the number of turns in the secondary winding .

$$\frac{N_P}{N_S} = \frac{320\ turns}{80\ turns} = \frac{4}{1}$$



Example A transformer has 800 turns on the primary and a turns ratio of 4. How many turns are on the secondary? 200

Transformer - Calculations

- Secondary Voltage/current) is determined by the primary voltage/current and turns ratio

Turns Ratio (n): $\boldsymbol{\frac{N_P}{N_S} = \frac{V_P}{V_S}}$ $\qquad V_S = V_P \frac{N_S}{N_P}$

- **Power Conservation:** Pp=Ps

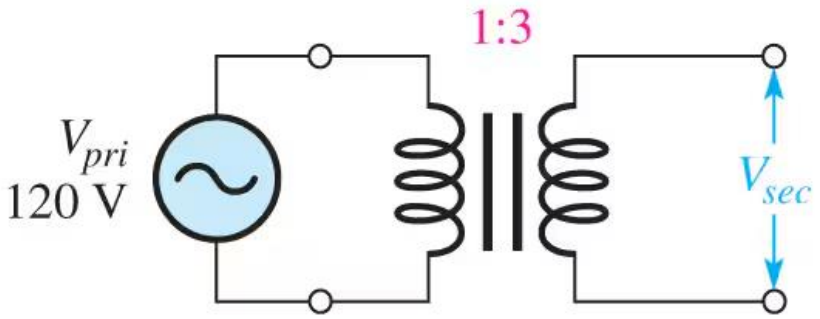
$$I_P V_P = I_S V_S \Rightarrow I_S = I_P \frac{V_P}{V_S}$$

$$\frac{I_S}{I_P} = \frac{N_P}{N_S} \Rightarrow I_S = I_P \frac{N_P}{N_S}$$

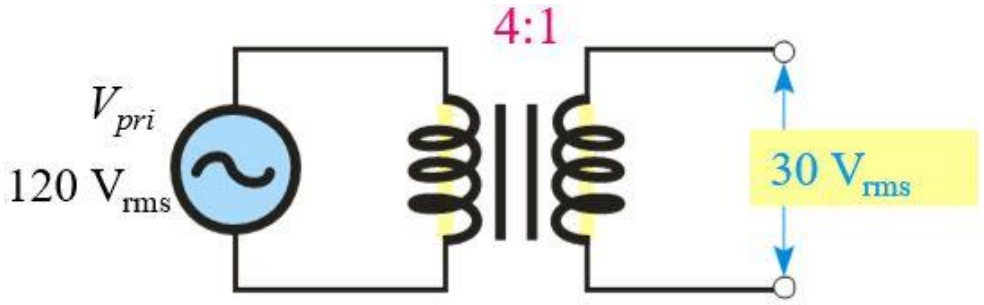
Step-up and Step-down transformers

- In a **step-up transformer**, the secondary voltage is greater than the primary voltage and $n < 1$.
- In a **step-down transformer**, the secondary voltage is less than the primary voltage and n >1.

Turns Ratio (n): $\frac{N_P}{N_S} = \frac{V_P}{V_S}$



n = 1/3
Vsec = 120/(1/3) = 360V



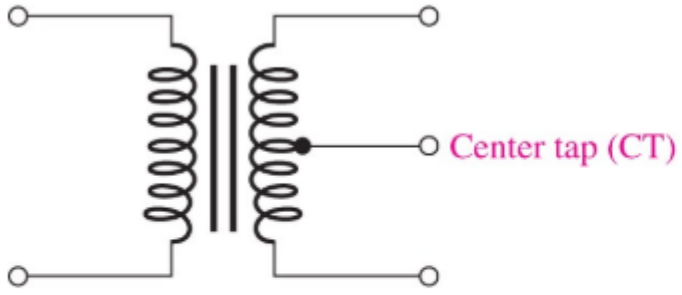
N = 4/1 = 4
Vsec = 120/(4) = 360V

Tapped and Multiple-Winding Transformers

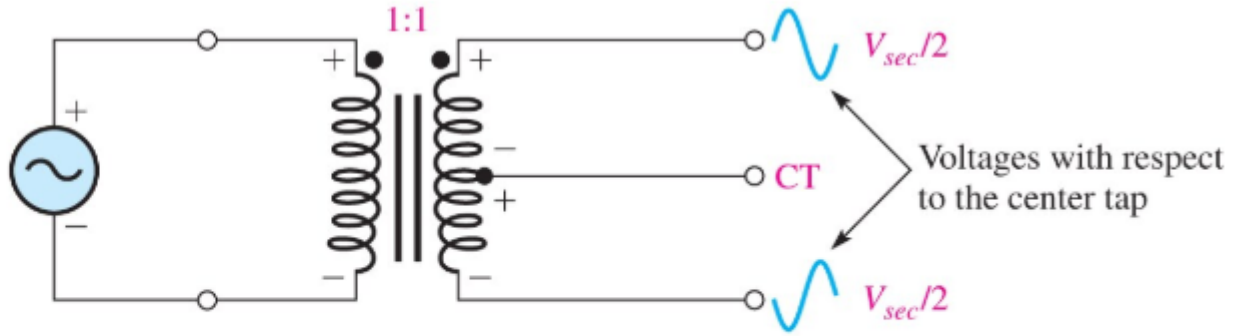
Transformation Variations:

- Center-tap transformer (tapped transformer)
- Multiple-winding transformer

Tapped Transformer: Frequently, it is useful to tap a transformer to allow for a different reference or to achieve different voltage ratings, either on the primary side or the secondary side.

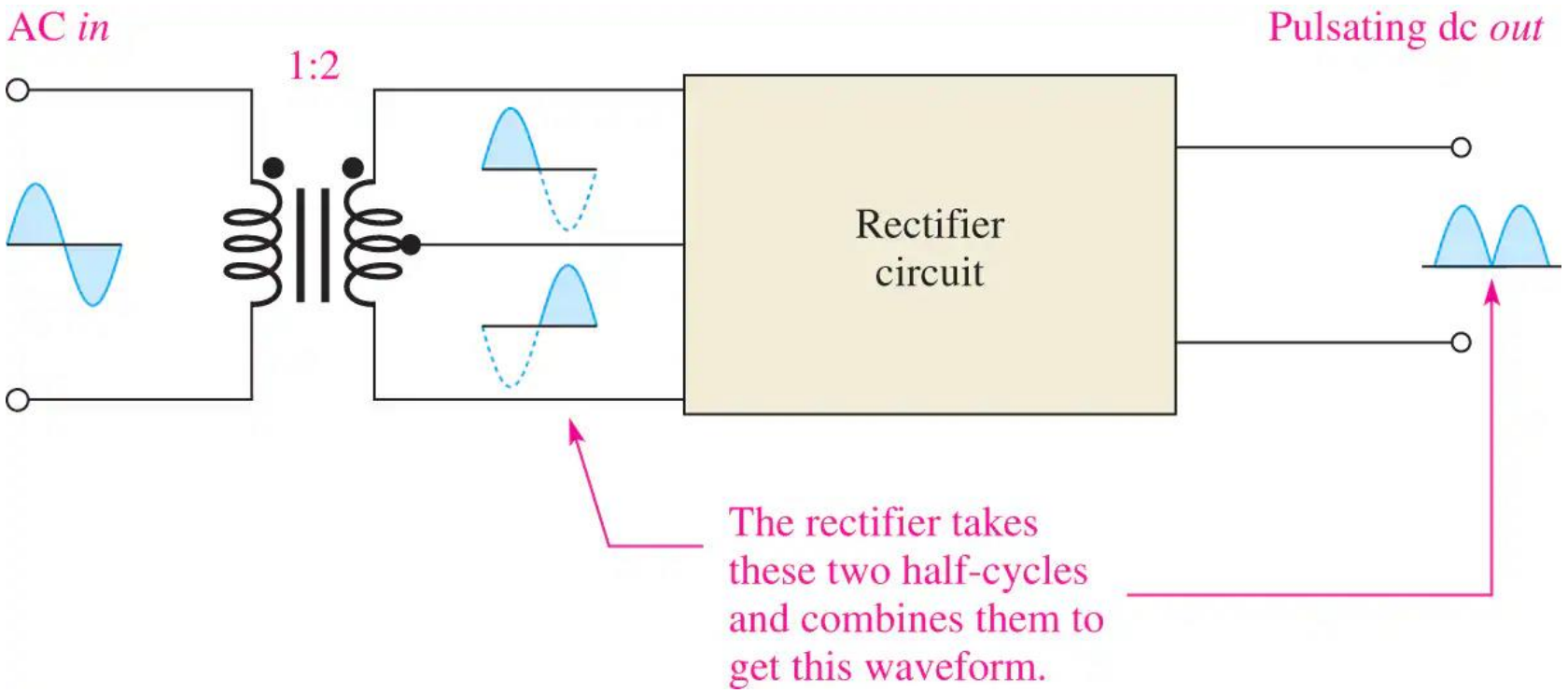


(a) Center-tapped transformer



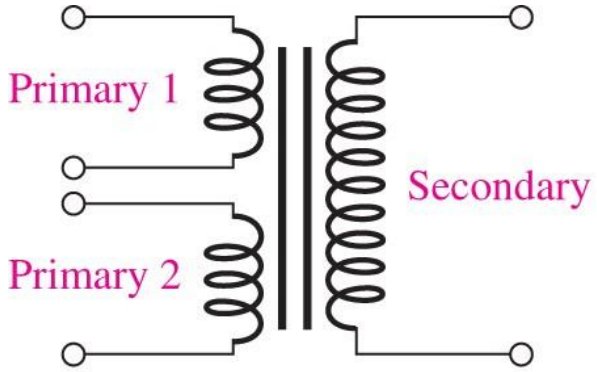
(b) Output voltages with respect to the center tap are 180° out of phase with each other and are one-half the magnitude of the secondary voltage.

Application of a center-tapped transformer in ac-to-dc conversion

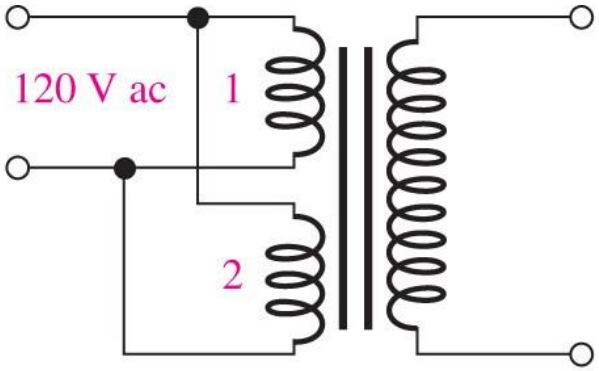


Multiple-Winding Transformers

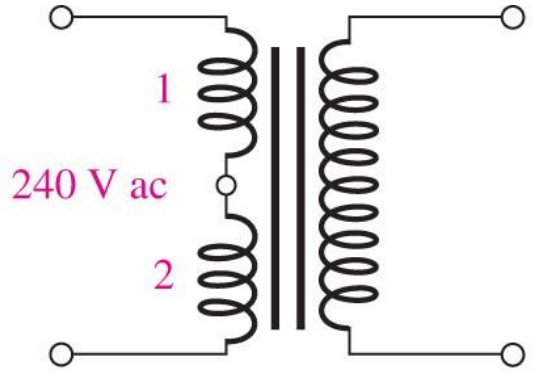
Multiple winding Transformer: can be on either the primary or secondary side. One application for multiple windings is to connect to either 120 V or 240 V operation.



(a) Two primary windings



(b) Primary windings in parallel for 120 V ac operation



(c) Primary windings in series for 240 V ac operation